

5.2 Uses of IoT in Different Areas

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 06 of 12

5.2 Uses of IoT in Different Areas

LEARN IT

IoT **improves efficiency** (doing more with less effort), **saves time**, and **enhances the overall quality of life**. It plays a significant role in developing modern **infrastructure** (the basic systems of a country — roads, power, water, hospitals). The source lists six areas.

1. Smart Cities

- **Traffic management:** Sensors adjust traffic signals automatically to reduce congestion (crowding). Example: signals on a busy Karachi road stay green longer for the direction with more cars.
- **Street lighting:** Lights turn ON/OFF based on daylight or movement, saving electricity.
- **Parking:** Sensors guide drivers to empty parking spaces, e.g. in a large shopping mall car park.
- **Waste management:** Smart bins send alerts when full, so trucks collect only full bins instead of driving to every bin.

2. Industrial Automation

- Sensors check machine **temperature and vibration** to prevent breakdowns (called predictive maintenance).
- Automated machines perform repeated tasks precisely.
- Quality-control sensors check products during manufacturing — for example, in the **textile industries of Sindh**, cameras and sensors spot faults in cloth as it is woven.

3. Healthcare (Smart Health)

- **Smartwatches** continuously measure heart rate and vital signs (pulse, oxygen level).
- **Remote monitoring devices** send patient data directly to doctors — vital for **rural areas of Pakistan**, where a specialist may be hundreds of kilometres away (e.g. a heart patient in Tharparkar monitored by a doctor in Karachi).
- **Emergency alert devices** inform hospitals instantly, for example when an elderly person falls.

4. Education (Smart Learning)

- **Smart classrooms:** digital boards and connected projectors.
- **Attendance systems:** biometric automated attendance for students and teachers (used in many Sindh government schools and colleges).
- **Resource management:** automated classroom lighting and AC control — off when the room is empty.

5. Home Automation

- **Smart lighting:** controlled remotely via mobile apps.
- **Security:** real-time monitoring via Wi-Fi cameras — you can watch your house gate on your phone.
- **Voice control:** using assistants (like Alexa or Google Assistant) to control TVs and appliances.

6. Smart Farming

Smart farming is **crucial for agriculture in Pakistan** (especially Sindh and Punjab) to **save water** and **improve yields** (the amount of crop produced).

- **Soil moisture sensors:** check if crops actually need water.
- **Automated irrigation:** turns water pumps ON/OFF based on soil data.
- **Drones:** monitor crop health from above, spotting disease or dry patches.

Smart farming process:



WORKED EXAMPLE

A cotton farmer near Nawabshah uses smart irrigation:

Step 1. A soil moisture sensor in the field reads 18% moisture.

Step 2. It sends the reading over the 4G network to a cloud service.

Step 3. The cloud AI checks: cotton needs at least 25% moisture, and the weather forecast shows no rain → decision "Needs water: YES".

Step 4. The actuator (pump switch) turns the tube-well ON.

Step 5. When moisture reaches 30%, the AI decides "enough" and the pump turns OFF.

Result: water is used only when needed — important because Sindh faces water shortages — and the farmer sees everything on his phone app.

REMEMBER IT

Remember the six areas with "**CHIEF H**": **C**ities, **H**ealth, **I**ndustry, **E**ducation, **F**arming, **H**omes.

COMMON MISTAKE

Mistake: Giving only the name of the area ("smart cities") in an exam answer. **Fix:** Always add *what* IoT does there — e.g. "smart bins send alerts when full" — and, if possible, a Pakistani example.

اردو میں سمجھیں

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔ اسمارٹ شہروں میں ٹریفک سگنل اور اسٹریٹ لائٹس خود کار ہوتی ہیں، اسپتالوں میں مریضوں کا ڈیٹا براہ راست ڈاکٹر تک پہنچتا ہے، اور کھیتوں میں مٹی کی نمی کے سینسر بتاتے ہیں کہ فصل کو پانی چاہیے یا نہیں، جس سے سندھ اور پنجاب میں پانی کی بچت ہوتی ہے۔

CHECK YOURSELF

1. How do smart bins help waste management?
2. Why is remote health monitoring especially useful in Pakistan?
3. In smart farming, which device is the actuator?

Answers: 1. They send alerts when full, so only full bins are emptied. 2. Rural areas lack doctors; devices send patient data directly to doctors far away. 3. The water pump (its switch).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Smart cities use IoT for:

- a. Traffic, waste, lighting
- b. Only K-maps
- c. Only sprints
- d. Only Access reports

MCQ 2. Health IoT example:

- a. Remote heart-rate monitor
- b. Bubble sort
- c. Primary keys
- d. Figma frames

MCQ 3. Farms use IoT to:

- a. Water when soil is dry
- b. Compile C
- c. Draw ER
- d. Write OSI

MCQ 4. Homes: a smart lock is IoT if:

- a. It is on the network and reports/acts
- b. It is a mechanical padlock only
- c. It is a Python tuple
- d. It is a pie

MCQ 5. Education IoT:

- a. RFID attendance, smart boards
- b. Only textbooks
- c. Only NAND labs
- d. Only waterfalls

MCQ 6. A risk in all these areas:

- a. Privacy / hacked devices
- b. Too much paper
- c. Too few charts
- d. Too much Big O

Extra short questions

1. Four application areas of IoT from the book.
2. One benefit in health.
3. One farm sensor + actuator pair.
4. Why security matters more as IoT grows.

Extra long question

1. Pick city, hospital, farm. For each: one device, one data item, one action, one risk.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Cities, health, farms, homes, industry, education — any four.
2. Doctors see vitals without the patient travelling.
3. Moisture sensor + water pump.
4. More doors into private spaces (cameras, locks).

Long

1. City: traffic cam / count / signal / surveillance. Hospital: SpO2 / alert nurse / false alarm. Farm: moisture / pump / overwater if sensor fails.